Dogzilla STM32 serial port communication protocol

Software interface

Standard TTL serial communication is adopted.

Baud rate	115200
Data bit	8
Stop bit	1
Parity bit	NONE

Data frame format

The data frame adopts a fixed format: frame header + frame length + data + checksum + frame tail.

frame header	Fixed as:0x55 0x00
frame length	**Bytes of the entire data frame**
data	**It has different meanings according to the instruction type, see Part II**
checksum	**Add the length and all bytes of the data, take the lowest byte, and then reverse it**
frame tail	**Fixed as:0x00 0xAA**

Instructions

Write，NO ACK（0x01）

header	length	type	First address:	Data	checksum	end
0x55 0x00		0x01		data		0x00 0xAA

The write command will continuously modify the data starting from the first address without generating a response.

>For example, modify the forward speed of the robot dog, the forward speed address is 0x30, and make it advance at the maximum speed, that is, the speed content is 0xff. The specific instructions are as follows:
>0x55 0x00 0x09 0x00 0x30 0xFF 0xC7 0x00 0xAA
>The calculation process of checksum is as follows::
>0x09+0x00+0x30+0xFF=0x138，Take the lowest byte 0x38，Reverse 0xC7

Read，ACK（0x02）

header	length	type	First address	Read length	checksum	end
0x55 0x00		0x02		uint_8		0x00 0xAA

The write command will continuously read the data from the first address without generating a response.
The format of the returned data packet is:

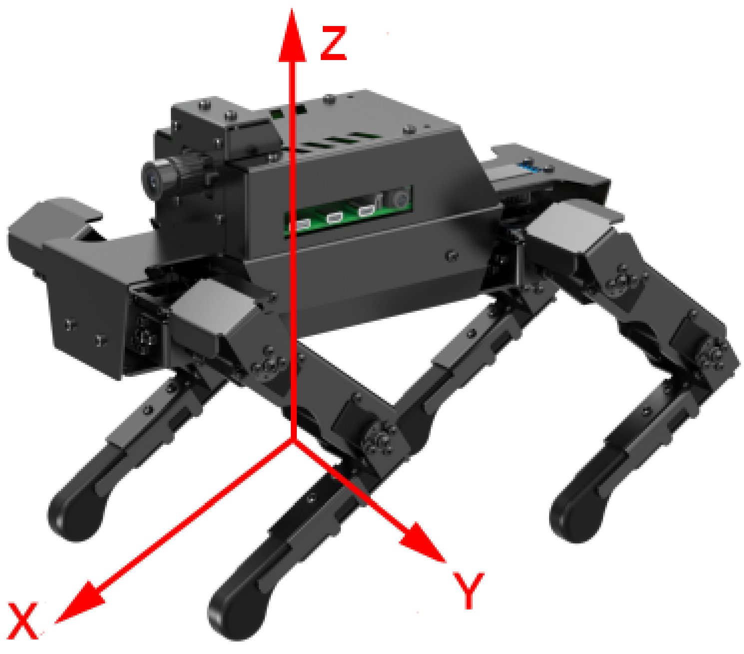
header	length	type	First address	data	checksum	end
0x55 0x00		0x12		data		0x00 0xAA

>For example, read the angles of 12 steering gears, 0x50 is the address of the first steering gear position, and 0x0c means to read 12 steering gears continuously. The specific instructions are as follows:
>0x55 0x00 0x09 0x02 0x50 0x0C 0x98 0x00 0xAA
>The calculation process of checksum is as follows:
>0x09+0x02+0x50+0x0C=0x67，Reverse 0x98
>Read return packet:
>0x55 0x00 0x14 0x12 0x50 0x80 0x80 0x80 0x80 0x80 0x80 0x80 0x80 0x80 0x80 0x80 0x80 0x89 0x00 0xAA

Robot dog coordinate system definition

Complete machine coordinate system

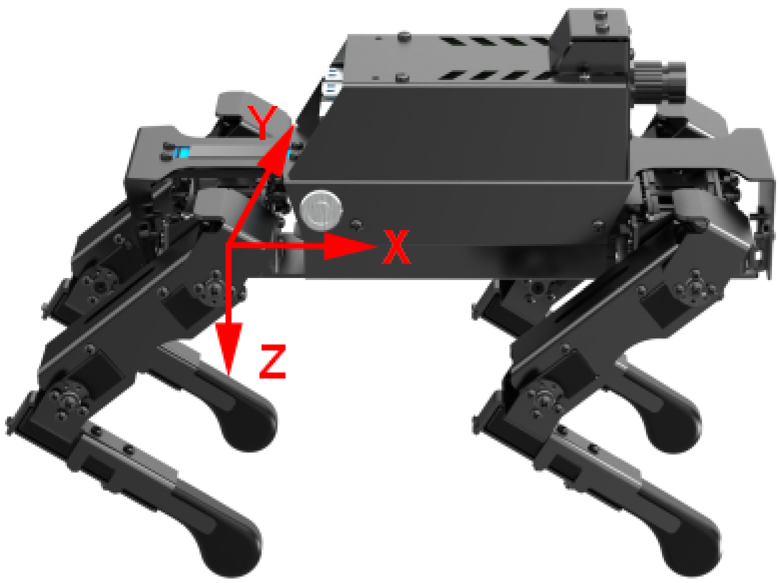
In the case of initial standing posture, the origin is directly below the fuselage. The forward direction of the robot dog is the x-axis forward direction, the left is the y-axis forward direction, and the upper is the z-axis forward direction.



Schematic diagram of coordinate system of robot dog

Single leg coordinate system

The single leg coordinate system is used to describe the position of the foot end. The four legs correspond to four independent single leg coordinate systems. The thigh joint is the origin, the forward direction of the robot dog is the x-axis forward direction, the left is the y-axis forward direction, and the lower is the z-axis forward direction.



Schematic diagram of robot dog single leg coordinate system

Robot dog memory table

addr	function	read/write	Init value	remarks	mode
0x00	working condition	read	0x00		status information
0x01	battery level	read	0xff	The range is 0-100, and the linear corresponding electric quantity is from the lowest value to the highest value	
0x03	Performance mode	write	0x00	0x00:Normal control mode	0x01:Cycle
0x04	Calibration mode	write	0x00	0x01:Enter calibration mode 0x00:Exit the calibration mode and complete the calibration	
0x20	Unloading steering gear	read/write	0x00	0x00:The steering gear is in normal working condition, 0x01:Unload all steering gear, 0x11-0x14:Unload 1-4 legs in sequence 0x21-0x24:Recover 1-4 legs in turn	Debug mode
0x21	Reset the zero position of the steering gear	write	0x00	0x00:The steering gear is in normal working state 0x01. The current position of all steering gear records is zero. After this register is set to 0x01, it will automatically jump to 0x00	

addr	function	read/write	Init value	remarks	mode
0x30	Forward and backward movement speed	read/write	0x80	The range is:0x00-0xff，Linear corresponding reverse maximum -forward maximum, positive direction according to robot coordinate system,The same below (note ③)	Machine mode
0x31	Left and right movement speed	read/write	0x80		
0x32	Clockwise and counterclockwise rotation speed	read/write	0x80	The direction follows the right-hand principle	
0x33	Body translation distance in X direction	read/write	0x80	The contact point between the foot end and the ground does not change but the body twists, the same below	
0x34	Body translation distance in Y direction	read/write	0x80		
0x35	Body height	read/write	0x80		
0x36	Body rotation angle about X axis	read/write	0x80	The direction follows the right-hand	
0x37	Body rotation angle about y axis	read/write	0x80		
0x38	Body rotation angle about z axis	read/write	0x80		

addr	function	read/write	Init value	remarks	mode
0x39	Rotate the body around the x-axis at a certain period	read/write	0x00	0x00:stop ,0x01-0xff:Linear corresponding min max rotation speed,This function and the direct setting position register cannot work at the same time, the same below	
0x3A	Rotate the body around the y-axis at a certain period	read/write	0x00		
0x3B	Rotate the body around the z-axis at a certain period	read/write	0x00		
0x3C	Stand still	read/write	0x00	0x00:stop,0x01-0xff:Linear corresponding min max tread height	
0x3D	Motion mode	read/write	0x00	0x00:Constant speed motion 0x01:Slow motion 0x02:High speed movement	
0x3E	action command	write	0x00	See action instruction table for details 0:Default stance 1-N:For each action (see right)	
0x80	Translational motion along the x-axis direction with a certain period	read/write	0x00	0x00:stop,0x01-0xff:Corresponding to the minimum maximum rotation speed, the movement range is half of the position limit	

addr	function	read/write	Init value	remarks	mode
0x81	Translational motion along the y-axis direction with a certain period	read/write	0x00		
0x82	Translational motion along the z-axis direction with a certain period	read/write	0x00		
0x40	Foot end position of left front leg in X direction	read/write	0x80	The range is 0x00-0xff, and the linear corresponds to the maximum value in the reverse direction - the maximum value in the forward direction. The positive direction is according to the robot coordinate system, the same below	Single leg mode
0x41	Foot end position of left front leg in Y direction	read/write	0x80		
0x42	Foot end position of left front leg in Z direction	read/write	0x80		
0x43	Foot end position of right front leg in X direction	read/write	0x80		
0x44	Foot end position of right front leg in Y direction	read/write	0x80		
0x45	Foot end position of right front leg in Z direction	read/write	0x80		

addr	function	read/write	Init value	remarks	mode
0x46	Foot end position of right hind leg in X direction	read/write	0x80		
0x47	Foot end position of right hind leg in Y direction	read/write	0x80		
0x48	Foot end position of right hind leg in Z direction	read/write	0x80		
0x49	Foot end position of left hind leg in X direction	read/write	0x80		
0x4A	Foot end position of left hind leg in Y direction	read/write	0x80		
0x4B	Foot end position of left hind leg in Z direction	read/write	0x80		
0x50	Position of steering gear with ID 11	read/write	0x80	The range is:0x00-0xff,Linear corresponding reverse maximum - forward maximum, the same below	Steering gear mode
0x51	Position of steering gear with ID 12	read/write	0x80		
0x52	Position of steering gear with ID 13	read/write	0x80		
0x53	Position of steering gear with ID 21	read/write	0x80		
0x54	Position of steering gear with ID 22	read/write	0x80		

addr	function	read/write	Init value	remarks	mode
0x55	Position of steering gear with ID 23	read/write	0x80		
0x56	Position of steering gear with ID 31	read/write	0x80		
0x57	Position of steering gear with ID 32	read/write	0x80		
0x58	Position of steering gear with ID 33	read/write	0x80		
0x59	Position of steering gear with ID 41	read/write	0x80		
0x5A	Position of steering gear with ID 42	read/write	0x80		
0x5B	Position of steering gear with ID 43	read/write	0x80		
0x5C	Set steering gear speed	read/write	0x80	The range is 0x00-0xff, and the linear corresponding minimum maximum value (only valid in this mode)	
0x5D	The position of the steering gear is set to stand	write	0x00	0x00:NONE 0x01:Restore standing position. After this register is set to 0x01, it will automatically jump to 0x00	
0x61	IMU state	read/write	0x00	0x00:close 0x01:Self stable mode	
0x62	ROLL angle	read	0x01		
0x63	PITCH angle	read	0x02		
0x64	YAW angle	read	0x03		